

LIL DITTO MVP — Engineering Specification

Version: 1.1

Status: Ready for Development

Document Type: Formal Engineering Specification

Brand Identity: Punk / Metal / Indie Rock

Core Hook: "SELFISH"

1. Product Overview

1.1 Objective

Deliver a high-performance, mobile-first, single-page application (SPA) that:

- Establishes the LIL DITTO brand identity
- Showcases audio, video, and visual media
- Captures fan emails with metadata tagging
- Provides a frictionless, high-impact experience optimized for social traffic

1.2 Primary KPI

- Email Conversion Rate (General_Fan + Merch_Interested)

1.3 Secondary KPIs

- Scroll depth
- Media engagement
- Social outbound CTR

2. System Architecture

2.1 Application Type

- Single Page Application (SPA)

2.2 Recommended Tech Stack

(Engineering may choose alternatives if requirements are met)

- Frontend: React or Vue
- Build: Vite or Next.js (static export mode)
- Hosting: Vercel / Netlify / Firebase Hosting
- Database: Mailchimp, SendGrid, or Firebase Firestore
- CDN: Cloudflare or Vercel Edge Network

2.3 Browser Support Matrix

Platform	Minimum Version
iOS Safari	14+
Android Chrome	95+
Desktop Chrome	95+
Desktop Safari	14+
Firefox	95+

2.4 Device Support

- Mobile: Required
- Tablet: Required
- Desktop: Required
- Ultra-wide monitors: Optional (centered layout acceptable)

3. Functional Requirements

Each section includes component definitions, states, and acceptance criteria.

3.1 Hero Section

3.1.1 Description

A full-screen vertical looping video with fallback for low-power or restricted autoplay environments.

3.1.2 Assets

- Video: H.265 + WebM
- Max file size: 5MB
- Aspect ratio: 9:16
- Fallback image: High-contrast PNG/WebP

3.1.3 Component: ``<HeroVideo />``

Props:

- srcVideoH265
- srcVideoWebM
- fallbackImage
- autoplay: boolean
- muted: boolean
- loop: boolean

States:

- loading
- playing
- fallback

3.1.4 Acceptance Criteria

- Video autoplays silently on all supported browsers.
- If autoplay fails → fallback image displays.
- Video loads within 1.0s on 4G.
- Text “SELFISH.” overlays at 90% viewport height.
- Landscape orientation shows letterboxed video.

3.2 Media Hub

3.2.1 Description

Embedded audio and video players with facade loading.

3.2.2 Components

`<AudioEmbed />`

- Spotify or Bandcamp embed
- Facade thumbnail until user interaction

`<VideoEmbed />`

- YouTube embed
- Facade thumbnail until user interaction

3.2.3 Acceptance Criteria

- No iframe loads until user clicks.
- Clicking facade replaces thumbnail with live embed.
- All embeds load asynchronously.
- If embed fails, show error state: "Media unavailable."

3.3 Notification Capture (Point 1: General)

3.3.1 Component: `<EmailCapture />`

Props:

- tag: "General_Fan"
- ctaText: "NOTIFY ME"
- successMessage: "YOU'RE IN."

3.3.2 Validation Rules

- Email must match RFC 5322 pattern.
- Reject disposable email domains (optional).

3.3.3 API Contract

POST /api/subscribe

Body:

```
{  
  "email": "string",  
  "tag": "General_Fan",  
  "timestamp": "ISO8601"  
}
```

Response:

- 200 OK → { "status": "success" }
- 400 → invalid email
- 409 → duplicate
- 500 → server error

3.3.4 Acceptance Criteria

- Submission occurs via AJAX.
- No page reload.
- Success message fades in within 300ms.
- Error messages appear inline.

3.4 About Section

Static content block.

Acceptance Criteria

- Text is selectable.
- Section loads instantly (no external dependencies).
- Typography matches brand spec.

3.5 Visual Pit (Gallery)

3.5.1 Component: ``<Gallery />``

Features:

- Masonry grid
- Lazy-loaded images
- WebP format required
- Lightbox on desktop
- Swipe navigation on mobile

3.5.2 Acceptance Criteria

- Images load only when entering viewport.
- Lightbox supports ESC to close.
- Swipe threshold: 30px horizontal movement.
- Max image file size: 300KB.

3.6 Merch Teaser

Static teaser block.

Acceptance Criteria

- Distressed mock-up image loads under 200KB.
- Text “DROPPING SOON.” is visible above the fold on mobile.

3.7 Notification Capture (Point 2: Merch)

Same component as Point 1 but with different props.

Props:

- tag: "Merch_Interested"
- ctaText: "I'M IN"
- successMessage: "YOU'RE IN THE PIT."

Acceptance Criteria

- Same as Point 1.

- Must appear in footer.

4. Data Model

4.1 Email Record

```
{  
  "email": "string",  
  "tag": "General_Fan | Merch_Interested",  
  "source": "LIL_DITTO_MVP",  
  "timestamp": "ISO8601",  
  "userAgent": "string",  
  "referrer": "string"  
}
```

4.2 Validation

- Email required
- Tag required
- Timestamp auto-generated

5. Non-Functional Requirements

5.1 Performance

- Lighthouse Performance Score: ≥ 90
- FCP: $< 1.5s$ on 4G
- TTI: $< 2.5s$
- All images WebP
- All embeds lazy-loaded

5.2 Accessibility

- WCAG 2.1 AA
- Contrast ratio: 4.5:1 minimum
- Focus states visible

- All images require alt text
- Video requires “Pause” control in fallback mode

5.3 SEO

- Title: LIL DITTO — SELFISH
- Meta description provided
- OG image: 1200x630
- Robots.txt: allow all
- Sitemap.xml: single URL

6. Analytics Specification

6.1 Event Taxonomy

Email Submissions

event: "email_submit"

params:

tag: "General_Fan" | "Merch_Interested"

location: "top" | "footer"

Media Engagement

event: "media_play"

params:

type: "audio" | "video"

source: "spotify" | "bandcamp" | "youtube"

Scroll Depth

event: "scroll_depth"

params:

percent: 25 | 50 | 75 | 100

Outbound Clicks

event: "social_click"

params:

platform: "instagram" | "tiktok"

7. Error Handling Matrix

Scenario	User Feedback	Logging
Invalid email	Inline error	Warn
Duplicate email	"Already subscribed"	Info
API offline	"Try again later"	Error
Network offline	Browser-native offline state	Warn
Embed fails	"Media unavailable"	Error

8. Risks & Dependencies

- Autoplay restrictions on iOS
- YouTube/Spotify embed rate limits
- CDN caching invalidation
- Asset delivery delays from creative team

9. Definition of Done

A feature is complete when:

- All analytics events fire correctly
- All accessibility checks pass
- All assets are optimized
- Lighthouse score ≥ 90
